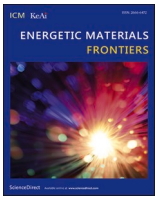
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Research Paper

A post-combustion reaction model for aluminum in thermobaric effects field

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ABSTRACT

The post-combustion effect is recognized as a critical factor responsible for the sustained high temperature and slow pressure decay in the blast field of thermobaric explosives. Focusing on CL-20 based thermobaric explosives, this study elucidates the underlying mechanism of the post-combustion effect through systematic experimental and theoretical investigations, and explores effective strategies for its enhancement. High-speed thermography is employed to record fireball morphology and temperature evolution, while a blast overpressure testing system is used to measure and compare shockwave parameters of thermobaric explosives with different post-combustion characteristics. The results demonstrate that thermobaric explosives exhibit a unique volumetric explosion mechanism characterized by a concentrated fireball, in stark contrast to the fragmented combustion observed in TNT. A higher oxidizer content yields a higher initial fireball temperature. After 30 ms, the post-combustion reaction of excess metallic fuel becomes dominant, causing TBX5% to exceed TBX30%, which has a higher oxidizer content. Secondary post-combustion reaction analysis also reveals that TBX5% exhibits more pronounced post-combustion fireball features. After 60 ms, the post-combustion temperatures of the two thermobaric explosive systems converge. The proposed post-combustion reaction model indicates that a higher fireball temperature broadens the particle size range of reactive aluminum powder, thereby improving combustion completeness. Notably, the hotter fireball (3001 K) driven by elevated oxidizer concentration leads to near-complete (>95%) and rapid combustion of aluminum powder, thus clarifying the enhancement of the thermobaric effect. Shockwave parameters correspond well with fireball evolution. In the near field, TBX5% shows higher overpressure and impulse, while in the far field, the supplementary effect of post-combustion on shockwave parameters becomes evident.

1. Introduction

Thermobaric explosives are a class of composite explosives typically formulated with high-energy monomolecular explosives, metal powders or composite metal powders, oxidizers, and other additives [1–3]. These materials exhibit high energy density and significant destructive power, making them suitable for applications requiring superior detonation performance [4–8]. Valued for its remarkable detonation performance and comprehensive properties, Hexanitrohexaazaisowurtzitane (CL-20) is a high explosive widely used in defense and ordnance [9–11]. This has enabled the development of various CL-20-based thermobaric explosives, significantly enhancing their application potential [12–17].

The superior thermobaric performance of CL-20-based thermobaric explosives over conventional explosives stems directly from their distinctive composition and microstructure [18–21]. The detonation of CL-20-based thermobaric explosives triggers a rapid reaction between CL-20 and the oxidizer, resulting in the creation of an extreme thermal and pressure environment concurrent with the production of abundant oxygen-containing species. This is followed by the combustion of the metallic fuel components, which leverages the released oxygen-containing species to further enhance the development of the high-temperature and high-pressure field, ultimately generating the thermobaric effect [22–27]. Consequently, optimizing the thermobaric performance of CL-20-based thermobaric explosives is critically dependent on the strategic selection and proportional optimization of the oxidizer and metallic fuel components. This work centers on the role

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of the oxidizer. Notable examples encompass ammonium perchlorate (AP), potassium perchlorate (KP), and ammonium dinitramide (ADN). Researchers have observed that thermobaric explosives incorporating AP or KP demonstrate notable breakthroughs compared to conventional explosives across a comprehensive set of properties, including thermal behavior, energy output, thermobaric effect, and fireball dynamics [28–34]. Nevertheless, a direct comparison between the two oxidants reveals that KP offers superior reactivity, enhanced chemical stability, lower hygroscopicity, and less toxic combustion products, rendering it more environmentally benign [35–39]. Consequently, KP was selected over AP for this investigation based on three principal considerations: firstly, AP-based thermobaric explosives produce distinct characteristic signals post-detonation, which is a well-documented phenomenon [40–43], secondly, AP possesses a lower density (1.95 g/cm^3) compared to KP (2.52 g/cm^3), and finally, the extensive existing research landscape concerning AP in thermobaric explosives [44–48] suggests that KP presents a comparatively less explored and potentially more fruitful avenue for novel investigation.

The oxidizer content in thermobaric explosives directly dictates its reactivity, which governs the initial thermomechanical environment upon detonation. This environment, in turn, critically affects the combustion efficiency and energy release of the metallic fuel, thereby determining the overall magnitude of the resultant thermobaric effect. Therefore, this study employs a broad variation in oxidizer loading to elucidate its impact on the thermobaric efficacy of CL-20 based thermobaric explosives.

2. Experimental

2.1. Raw materials

The materials utilized in this study included: CL-20 (industrial grade, ε-polymorph, Liaoning Qingyang Special Chemical Co., Ltd.); aluminum powder (industrial grade, Ansteel Industrial Micro Fine Aluminum Powder Co., Ltd.); potassium perchlorate (Industrial Type II, Jiangxi Yongning Technology Co., Ltd.).

2.2. Sample preparation

(1) CL-20 based thermobaric explosives: In order to investigate the influence of high oxidizer content on the afterburn fireball, the compression-molded CL-20 based thermobaric explosive, namely TBX30% and TBX5%, were designed. The component compositions of the two formulations are listed in Table 1. It can be observed that the content of aluminum powders and polymeric binder are both fixed, ensuring a consistent mass of the metal fuel responsible for the thermobaric effect in both formulations. Additionally, both formulations exhibit nearly identical and excellent formability. The variation in KP content is significant, where TBX5% and TBX30% correspond to an increase in KP content from 5% to 30%. Along with this change, the main explosive component in the formulation, CL-20, decreases from 53% to 28%. Due to the excellent initiation performance of CL-20, both thermobaric explosives can undergo complete detonation in the presence of Cyclotrimethylenetrinitramine-based (RDX-based) initiators.

(2) TNT charge: For comparative purposes, TNT charges with the same volume as TBX30% and TBX5% were fabricated via a melt casting method to serve as reference samples, designated as $Ref_{TNT}30\%$ and $Ref_{TNT}5\%$, respectively. Furthermore, theoretical calculations for the thermobaric explosive formulations were performed using the EXPLO5 software, [49] with the results summarized in Table 1.

2.3. The blasting area lay-out

The open-space static field blast tests were conducted for the thermobaric explosives and the reference TNT charges. The tested charge was located at the center of the field, at an elevation of 1 m above the ground, as depicted in Fig. 1. Beginning from the explosion center, two lines of wall pressure sensors are arranged at a 60° angle. Each line is equipped with four sensors, which are situated at distances of 2, 4, 6, and 8 m from the explosion center, respectively. The explosion events including the fire-ball duration and its dimensions were monitored by two infrared thermal imaging devices and one high-speed video recorder that positioned at a distance of 16 m from the explosion center, at the same horizontal level as the tested charges.

3. Results and discussion

3.1. The characteristic and evolution of the explosive fireball

The researchers conducted explosive fireball tests on the CL-20 based thermobaric explosive samples and TNT samples designed in Section

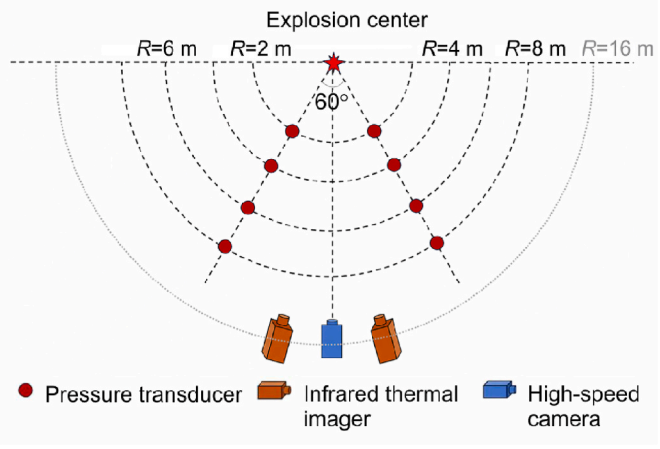


Fig. 1. Schematic of the experimental setup.

Table 1
Sample information and theoretical calculation results.

Formulations and Mass Data of the Explosive Charges					
Samples	Aluminum powder/wt%	CL-20/wt%	KP/wt%	Polymeric binder/wt%	Charge quality/g
TBX30%	36	28	30	6	1000
TBX5%	36	53	5	6	1000
Theoretical Calculation Results of Parameters for Thermobaric Explosive Samples					
	Density/g·cm^{-3}	Detonation heat/kJ·kg^{-1}	Detonation temperature/K	Detonation pressure/GPa	Detonation velocity/m·s^{-1}
TBX30%	2.37	9088.80	6163.48	26.56	6947.62
TBX5%	2.25	9303.10	5759.87	24.33	7340.70

2.2. The explosive fireballs were captured using a high-frame-rate (200 frames per second) infrared thermal imager, allowing for the observation of the explosive detonation and the expansion process of the detonation products, as well as the acquisition of the highest temperature point on the fireball surface and the trend of temperature variation. The temperature contour maps of the explosive fireballs, captured at specific time intervals, are illustrated in Fig. 2.

Fig. 2 shows the evolution process of the explosion fireball from formation to gradual dissipation for all samples. It can be seen from the figure that at the moment of fireball formation (5 ms), the explosive fireballs of both the two thermobaric explosive samples TBX30% and TBX5% and the two TNT samples $Ref_{TNT}30\%$ and $Ref_{TNT}5\%$ are in the shape of a pagoda or flying saucer, with the diameter of the fireball exceeding its height. At this point, the size of the explosive fireball has expanded to several dozen times that of the charge, exhibiting a horizontal expansion rate of approximately 38 to 45 times and a vertical expansion rate of about 20 to 30 times. For the samples tested in our experiment, the fireballs have expanded almost to its maximum size at 5 ms. After 5 ms, under the influence of reflected shock waves and airflow, the explosive fireballs began to drift, rise, and deform. Overall, both the horizontal and vertical dimensions of the fireballs slightly increased, but the high-temperature fireballs became more fragmented, leading to a more uneven temperature distribution within the fireballs. The dimensions of the explosive charge and the size of the explosive fireball at the 5 ms mark are detailed in Table 2.

At this juncture, a comparison between TBX30% and TBX5% shows that the horizontal dimension of the explosive fireball of TBX5% is 377.28 cm, which is larger than that of TBX30% (327.36 cm), while its vertical dimension is 197.76 cm, smaller than the latter's 250.56 cm. It is worth noting that the aspect ratios (charge height: charge diameter) of both types of explosive charges are close to 1:1, and that of TBX5% is even slightly higher than that of TBX30%. However, this more elongated charge structure does not impede the fireball's horizontal expansion, which likely results from the predominance of the monolithic explosive content.

3.2. The evolution trend of the peak temperature on the surface of the fireball

Peak temperatures on the surface of the explosion fireball are extracted from each frame of the infrared thermal imaging temperature cloud map, and the evolution of these peak temperatures is chronologically recorded. Considering that a fireball with a temperature too low to inflict effective damage, or one that requires a prolonged duration to cause such damage, is less relevant, the statistical analysis solely accounts for explosive fireballs with peak temperatures exceeding 1200 °C, and the results are shown in Fig. 3.

Fig. 3a illustrates the comparative curves of peak temperature between TBX30% and $Ref_{TNT}30\%$, and between TBX5% and $Ref_{TNT}5\%$, while Fig. 3b showcases the curves comparing the peak temperature between TBX30% and TBX5%. It is noteworthy that for thermobaric explosives, the first data recorded post-detonation represents the highest temperature point on the curve tracing the evolution of the surface temperature on the explosion fireball. This point is also the peak data in the temperature contour map at the 5 ms mark in Fig. 2. Subsequently, the peak temperature on the surface of the fireball generally exhibits a decreasing trend. Specifically, the peak temperature of TBX30% shows a steady decline, while that of TBX5% remains approximately constant within the range of 15 ms to 45 ms after detonation, before gradually decreasing. This phenomenon is subject to more in-depth discussion later on.

The peak temperature of on the surface of the fireball for $Ref_{TNT}30\%$ and $Ref_{TNT}5\%$ exhibits a pattern of an initial slight increase followed by a gradual decrease. Comparing TBX30% and TBX5% to $Ref_{TNT}30\%$ and $Ref_{TNT}5\%$ of the same charge volume, the greatest disparity in the fireball surface peak temperatures is observed at the onset of the

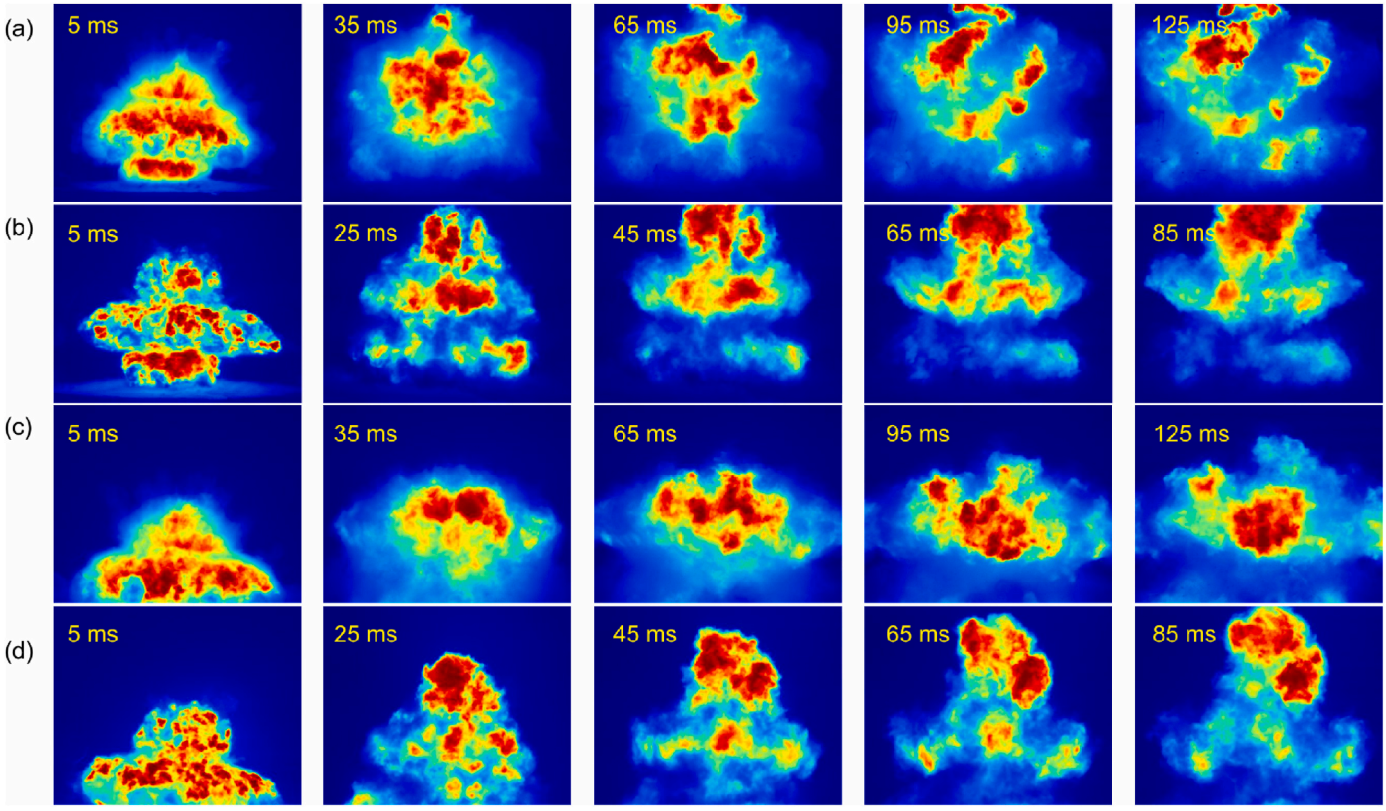


Fig. 2. Temperature contour maps of the CL-20 based thermobaric explosive sample and the TNT sample of the same volume: (a) TBX30%; (b) $Ref_{TNT}30\%$; (c) TBX5%; (d) $Ref_{TNT}5\%$.

Table 2
Data on Charge Dimensions, Fireball Size, and Expansion Rate of four samples.

Samples	Fireball diameter /cm	Fireball height /cm	Charge diameter /cm	Charge height /cm	Lateral dimensional expansion rate /%	Longitudinal dimension expansion rate /%
TBX30%	327.36	250.56	8.50	8.17	3851.29	3066.83
Ref_{TNT}30%	332.16	218.88	8.50	8.17	3907.77	2679.07
TBX5%	377.28	197.76	8.50	8.77	4438.59	2254.96
Ref_{TNT}5%	336.96	181.44	8.50	8.77	3964.24	2068.87

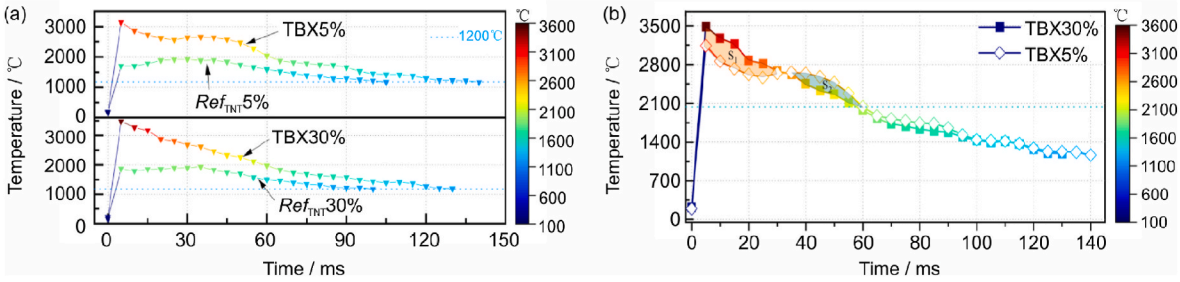


Fig. 3. Comparison of peak temperatures: (a) TBX30% *vs.* Ref_{TNT}30%, TBX5% *vs.* Ref_{TNT}5%; (b) TBX30% *vs.* TBX5%.

explosion. Taking TBX30% and Ref_{TNT}30% as examples, this temperature difference can exceed 1500 °C. As the fireball expands, the gap in peak temperatures between thermobaric and TNT charges gradually narrows. After about 100 ms post-detonation, the difference stabilizes at approximately 250 °C. This means when the peak temperature of the TNT explosion fireball falls to around 1200 °C, the peak temperature on the surface of the thermobaric explosion fireball remains above 1450 °C.

Fig. 3b displays the evolutionary curves of the peak temperatures on the surface of the fireball for TBX5% and TBX30%. This aspect constitutes one of the keys focuses of the study, namely the pattern of how the oxidizer content influences the high-temperature effects within the thermobaric effect. From the figure, it is evident that after 60 ms post-detonation, the two curves nearly converge, exhibiting almost identical evolutionary patterns. The primary distinction in the surface peak temperatures of TBX5% and TBX30% explosion fireballs lies within the 0 to 60 ms interval, a topic that is explored in detail subsequently. Within the scope of this discussion, the 30 ms mark represents a distinct turning point. Prior to this juncture, the surface peak temperature of the TBX30% explosion fireball surpasses that of TBX5%, however, post this point, the situation reverses. Typically, the explosion reaction of thermobaric explosives proceeds through three consecutive stages. In the initial stage, the high explosive and a portion of the oxidizer undergo a detonation reaction to generate a high-temperature, high-pressure environment. Subsequently, in the second stage, the metallic fuel in the formulation utilizes this environment to undergo an anaerobic exothermic reaction with the detonation products. Finally, in the third stage, the unreacted fuel participates in secondary post-combustion reactions with turbulent entrained air. We believe that at the conclusion of the second stage of the thermobaric explosive reaction, the majority of the oxidizer has either undergone detonation reactions or rapid decomposition. For TBX30%, due to its higher content of oxidizer, the explosive decomposition releases a substantial amount of heat along with a significant generation of oxidizing products. This increases the concentration of oxidizing substances in the environment during the anoxic combustion of aluminum powders, thereby enhancing the reaction extent of the aluminum powders in the second stage of the thermobaric explosive reaction. Therefore, during this stage, corresponding to the first 30 ms of the fireball temperature evolution curve, TBX30% exhibits a higher initial explosion fireball temperature.

This theory also accounts for the evolutionary trend of the peak temperature curve of the TBX5% explosion fireball. Specifically, the lower oxidizer content hinders the anoxic combustion of aluminum powders. During this stage, the reaction of aluminum powders is somewhat subdued, leaving a substantial amount unreacted for participation in the third stage of the thermobaric explosive reaction. Hence, the initial explosion fireball temperature of TBX5% is marginally lower than that of TBX30%. Nevertheless, this initial temperature is adequate to sustain the remaining aluminum powders in an aerobic combustion reaction with the secondary convective air, thereby maintaining a nearly constant peak temperature of the TBX5% explosion fireball in the 15 ms to 45 ms interval.

The aforementioned constitutes the rationale behind the observed differences in the peak temperature of the explosion fireballs of the two types of thermobaric explosives, with the 30 ms mark post-detonation serving as a turning point. As the fireball continues to expand, especially after 60 ms post-detonation, the exothermic chemical reactions within the explosion fireball struggle to compensate for the heat loss caused by convection or radiation, resulting in a gradual decrease in the peak temperature on the surface of the fireball.

3.3. *The secondary afterburn reactions analysis*

The secondary combustion characteristics of the thermobaric explosive samples were analyzed by processing infrared thermography data at the instant of maximum temperature and by capturing the spatiotemporal trajectory of the fireball's surface hotspot, as depicted in Figs. 4 and 5. Specifically, Fig. 4 presents a magnified view of the region near the maximum temperature point on the sample surface when an afterburning reaction occurs following the explosion, accompanied by sustained high-temperature combustion. In Fig. 5, the star symbols represent the locations of the highest temperature point on the fireball surface at different instants. These points are connected by straight lines to indicate the trajectory of the highest temperature point over time, while the circles denote the distinct stages of this temperature migration and the spatial span of the highest temperature point's movement.

The secondary afterburning behavior is elucidated from magnified infrared thermographic details in Fig. 4. The first column (Fig. 4a–c) shows enlargements of the original data, and the second column (Fig. 4d–f) provides modeled interpretations that depict the secondary combustion sequence. As can be observed from Fig. 4, TNT, TBX5%, and TBX30% all exhibit certain afterburning characteristics. Specifically, a phenomenon of re-ignition occurs in regions that had previously cooled after initial combustion, resembling a form of localized micro-explosion. Regarding the intensity of the afterburning reaction, TBX5% exhibits a more vigorous secondary combustion process. Notably, the surface hotspot of the explosion fireball is observed to be located precisely

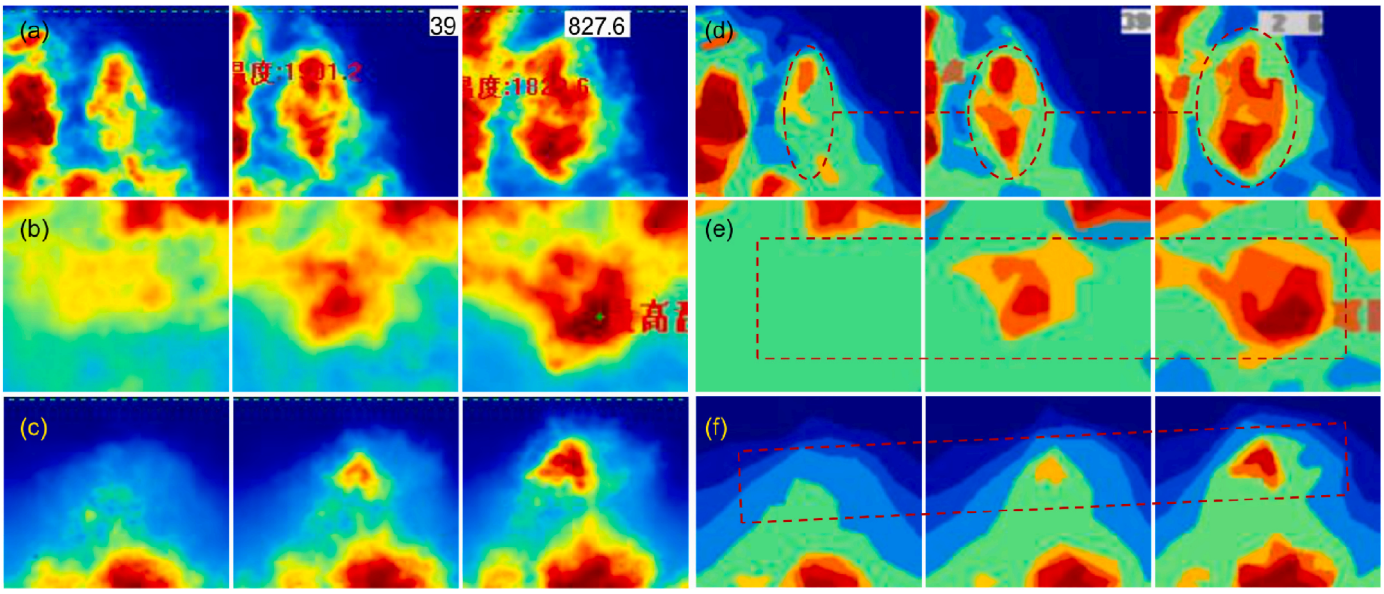


Fig. 4. Infrared thermography of the explosion fireballs: (a) (d) $Ref_{TNT}30\%$; (b) (e) TBX5%; (c) (f) TBX30%.

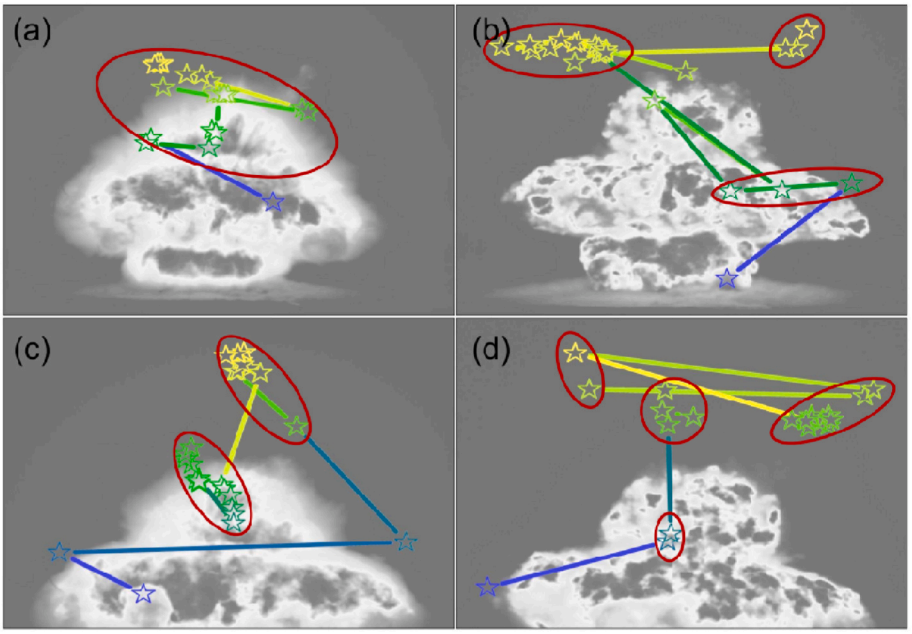


Fig. 5. Migration of the surface hotspot in the explosion fireballs: (a) TBX30%; (b) $Ref_{TNT}30\%$; (c) TBX5%; (d) $Ref_{TNT}5\%$.

within this secondary reaction zone. In comparison, the secondary afterburning intensity of TBX30% is slightly weaker than that $Ref_{TNT}30\%$. This might be attributed to the high oxidizer content in TBX30%, or it could merely represent an atypical scenario confined to the specific observation area.

In addition, the migration of the surface hotspot on the explosion fireball was discussed. By capturing the location of the maximum temperature point of the fireball at different times, Fig. 5 was obtained. It can be observed that the surface hotspot of the TBX30% and TBX5% explosion fireball is relatively concentrated within a smaller spatial region (Fig. 5a–c). This signifies that the thermobaric explosive exhibits a distinct volumetric explosion character. This volumetric behavior persists throughout all three stages of the explosion event. Benefiting from the effective coupling between the high-temperature/high-pressure blast field and the processes of energetic metal fuel dispersal, fresh air entrainment, and metal-air mixing, the secondary afterburning reaction largely maintains this volumetric nature. Furthermore, the most intense reaction is localized precisely at the three-dimensional spatial temperature peak within the blast field. TBX5% exhibits a more pronounced migration of its surface hotspot compared to TBX30%. This phenomenon is primarily due to the lower oxidizer content in TBX5%, which results in a lower initial temperature and pressure within the blast field relative to the formulation with higher oxidizer content. Consequently, the local environmental parameters in certain regions of the blast field are insufficient to support a sufficiently rapid and complete metal-air chemical reaction, thereby leading to the observed significant shift of the fireball's hottest point. This observation corroborates the conclusion that a higher oxidizer content promotes the oxygen-deficient combustion reactions in the second stage of the thermobaric explosion sequence, while simultaneously attenuating the secondary afterburning

reaction. The more extensive migration of the surface hotspot in the $Ref_{TNT}30\%$ and $Ref_{TNT}5\%$ fireball (Fig. 5b–d) serves as further evidence that a spatiotemporal mismatch between the high-temperature/high-pressure blast field and the distribution of combustible components can significantly hinder the secondary afterburning reaction. TNT detonation is typically incomplete, resulting in partially oxidized products. When air is entrained, secondary afterburning can still occur; however, by this time, the temperature of the initial blast field has already significantly decayed. Consequently, the most intense zone of the secondary reaction may not coincide with the location of the initial temperature peak, which is visually manifested in the figure as the large-scale migration of the fireball's surface hotspot.

3.4. Practical shock wave pressure-time curve

According to the layout method described in Section 2.3, PCB wall pressure sensors were used to measure the blast wave parameters of the explosives at distances of 2, 4, 6, and 8 m. The raw data of the blast wave parameters are listed in Table 3, the pressure-time history curves are shown in Figs. 6 and 7 presents a comparative analysis of the blast wave parameters for different samples, including blast wave overpressure, the product of pressure and time (i.e., impulse), and the positive pressure duration.

The blast wave parameter test results show that the overall measurements conform to expected patterns. The blast wave pressure of the thermobaric explosive samples exceeds that of the TNT samples at all distances. Comparison of the two TNT samples, $Ref_{TNT}30\%$ and $Ref_{TNT}5\%$, reveals that their pressure values are nearly identical at each distance, which is mainly because their qualities are roughly the same. However, when comparing the two thermobaric explosive samples, TBX30% and TBX5%, the latter exhibits significantly higher overpressure. This difference is attributed to the varying contents of the primary explosive in the two formulations: TBX5% contains 53% CL-20, whereas TBX30% contains only 28%. Consequently, TBX30% has a higher metal powder content, and its advantage lies primarily in post-combustion energy release. Therefore, at far-field distances (4 m and beyond from the detonation center), the advantages of the TBX30% sample in terms of blast wave impulse and positive pressure duration become evident. These parameters are generally higher than those of the TBX5% sample, which is a crucial manifestation of the supplementary effect of post-combustion on blast wave energy release. Thus, the ratio of primary explosive to metal powder content in thermobaric explosives plays a vital role in achieving superior thermodynamic or kinetic performance.

4. Numerical analysis

Based on the research findings presented in the preceding sections, this section performs theoretical calculations on the degree of combustion of aluminum powder within the fireball through modeling and simulation.

4.1. Theoretical hypothesis

Depending on the particle size, aluminum powders can either contribute to the detonation pressure and the blast wave by burning behind the strong convective detonation flows, or they can reaction in the secondary afterburn fireball, leading to the generation of thermobaric effects. With regards to the micron-sized aluminum powders utilized in this study, their larger particle size and lower reaction rate lead to an extensive combustion duration that significantly exceeds the timescale of the detonation. Consequently, the predominant mode of reaction is characterized by secondary afterburn. The combustion of aluminum particles within the fireball is primarily influenced by the particle size of aluminum and the temperature of the fireball. Theoretical calculations of the extent of aluminum combustion require the following assumptions.

Initially, the combustion of aluminum powder is modeled as a gas-phase reaction process. Generally, before reaching the melting point, aluminum powders undergo a low-temperature oxidation stage, following the diffusion oxidation mechanism (DOM). For thermobaric explosives, the temperature within the fireball often remains higher than the aluminum's boiling point (2740K) until the completion of its expansion, resulting in gas-phase combustion at this stage. Subsequently, it is postulated that the alumina shell undergoes instantaneous fracture and spallation under the impact of the detonation shock wave. The results demonstrate that under detonation shockwave loading, the rupture response times of the alumina shell for aluminum powders with particle sizes of 0.1 μm, 2 μm, 10 μm, and 20 μm are 0.02 ns, 1.3 ns, 3.3 ns, and 5.4 ns, respectively. This time scale is significantly shorter than the combustion time scale of the aluminum powders, rendering this stage negligible. Secondly, it is assumed that the detonation system is in a state of thermal equilibrium. That is to say, the heat absorbed by aluminum powders as their temperature rises to match that of the detonation products, as well as the heat released during aluminum combustion, are disregarded. Simultaneously, it is assumed that following the completion of the detonation reaction, the aluminum particles and the detonation products rapidly equilibrate to a common temperature. Finally, it is assumed that the reaction between gaseous aluminum and the oxidizing detonation products is instantaneous.

4.2. Establishment of the model

The combustion rate of aluminum powder, under the stated assumptions, is primarily determined by the gasification rate of reactive aluminum and the diffusion rate of products. The former is formulated using the Boltzmann law, with the latter being simplified to a function of the diffusion coefficient α only. Under these premises, the aluminum combustion equation is derived as follows:

$$C_c = 1 - \left(1 - \propto \frac{\sqrt{3mkt}}{2\pi r^3 \rho^R} e^{\frac{E}{RT_t}}\right)^3$$

Where *Cc* represents the combustion degree of aluminum particles, α is the diffusion coefficient, $\boldsymbol{m}$ is the mass of aluminum atoms, g·mol^{-1}, $\boldsymbol{k}$ represents the Boltzmann constant, 1.38×10^{-23} J K^{-1}, $\boldsymbol{T}$ represents the system temperature in Kelvin, $\boldsymbol{r}$ is the radius of aluminum atoms, m, $\boldsymbol{\rho}$ is the density of aluminum atoms, g/m^3, $\boldsymbol{R}$ is the ideal gas constant, 8.314 J mol^{-1} K^{-1}, $\boldsymbol{E_v}$ represents the gasification enthalpy of aluminum, 287 kJ/mol, $\boldsymbol{t}$ is the combustion time, s.

Table 3
Original data of shock wave parameters for the samples.

	2m			4m			6m			8m		
	p / kPa	I / kPa	Δt_+ / ms	p / kPa	I / kPa	Δt_+ / ms	p / kPa	I / kPa	Δt_+ / ms	p / kPa	I / kPa	Δt_+ / ms
TBX30%	555.46	0.19	1.22	114.14	0.10	2.58	111.71	0.13	3.38	17.05	0.03	4.01
$Ref_{TNT}30\%$	443.10	0.16	1.28	88.65	0.07	2.67	87.38	0.10	3.15	14.90	0.02	3.28
TBX5%	619.06	0.25	1.40	128.55	0.09	2.55	114.86	0.11	3.23	19.43	0.02	4.61
$Ref_{TNT}5\%$	432.50	0.16	1.12	99.73	0.08	2.93	86.48	0.10	3.18	14.90	0.02	3.26

Note: The physical symbol p represents shock wave overpressure, I represents specific impulse, and Δt_+ represents positive pressure duration.

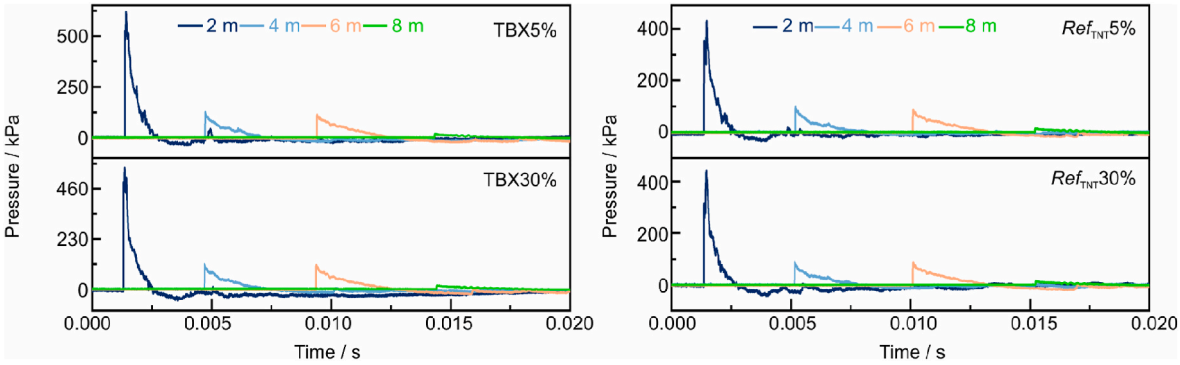


Fig. 6. The pressure-time history curves of different samples.

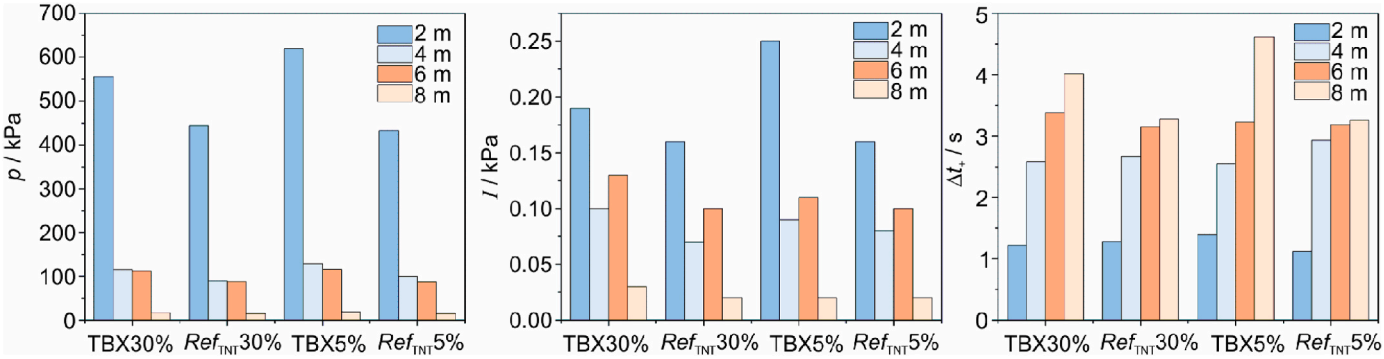


Fig. 7. Comparison and analysis of the shock wave parameters of the samples.

When $C_c = 1$, the burn-off time of aluminum particles (t_c) can be determined through solving the equation, namely

$$t_c = \frac{2\pi r^3 p^R}{\propto\sqrt{3mkte}^{-\frac{E_v}{RT}}}$$

The equation for the burn-off time illustrates that the duration of aluminum particle combustion is solely dependent on the particle size of the aluminum and the ambient temperature. In this study, the micron-sized aluminum powders employed exhibit a particle size distribution pattern as depicted in Fig. 8a, while the ambient temperature corresponds to the fireball temperature. Taking into account the limitations imposed by the sampling frequency, the fireball temperature isn't represented as a continuous curve, but rather as a sequence of discrete points. Continuously calculating the combustion degree presents considerable difficulties and substantial challenges. Hence, we employ two simplified models—the Lower Temperature Bound (LTB) Model and the Average Fireball Temperature (AFT) Model—to examine the combustion behavior of aluminum powders within the TBX5% and TBX30% [50,51].

4.3. Results and discuss

Two simplified models were used separately to analyze the combustion behavior of aluminum powders in the TBX30% and TBX5% samples. The results are shown in Fig. 8.

The analysis was performed using the Lower Temperature Bound (LTB) Model (Fig. 8a).

Given that the fireball temperature falls below the melting point of alumina, a shift occurs in the diffusion behavior of the combustion products. Consequently, the previously mentioned aluminum particle combustion equation may not be entirely applicable. Consequently, utilizing the melting point of alumina as the lower temperature bound, we compute the extent of aluminum powders combustion for temperatures surpassing this point. Fig. 3b illustrates the fireball temperature curves for TBX5% and TBX30%, which converge after 2327K. Fireballs with temperatures exceeding 2327K exhibit a consistent duration of approximately 60 ms, corresponding to the reaction zone within the burn-off time curve depicted in Fig. 8a. At this juncture, it becomes evident that aluminum powders with diameters less than 33.4 μm undergo complete reaction. According to the particle size distribution curve, the aluminum powders achieving full reaction comprise 54.22% of the total aluminum mass. Based on the trend of the burn-off time, the estimated time required for complete combustion of all aluminum powders is approximately 1.6 s. Evidently, this model diverges significantly from common understanding; the extent of aluminum powders combustion is notably underestimated, while the total combustion duration is considerably overestimated. This is primarily due to the fact that the LTB model entirely overlooks the high-temperature zone of the secondary afterburn fireball in thermobaric explosives. While it might be applicable to conventional aluminized explosives, it falls short in accurately describing the afterburning process of thermobaric explosives.

The analysis was performed using the Average Fireball Temperature (AFT) Model (Fig. 8b and c).

As the Lower Temperature Bound Model notably underestimated the response of aluminum powders within a high-temperature fireball, we adopted the mean value between the maximum temperature of the fireball and the melting point of alumina as the average fireball temperature for a reevaluation of the reaction properties of aluminum powders. Fig. 8b and c depict the burn-off times and the aluminum reaction zones for TBX5% and TBX30%, respectively. According to this model, the mean fireball temperature for TBX5% measures 2863K, while that for TBX30% registers at 3001K. Observations reveal that, under conditions of 2863K and 3001K, the burn-off duration for aluminum powders stands at 88 ms and 43 ms, respectively. When juxtaposed with the LTB model, the temporal dimension of the reaction zone experiences

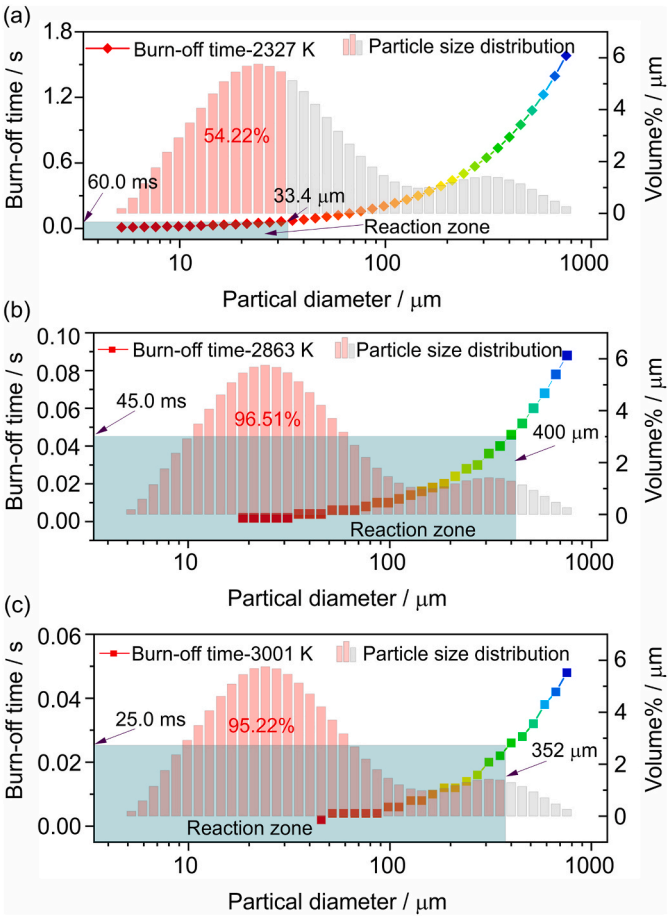


Fig. 8. Combustion time and reaction zone:(a) Calculation results obtained using the Lower Temperature Bound Model, where the results for both samples are consistent; (b) Calculation results for TBX5% using the Average Fireball Temperature Model; (c) Calculation results for TBX30% using the Average Fireball Temperature Model.

compression within the AFT model, resulting in a reduced duration for aluminum powders combustion in the high-temperature fireball. Specifically, at 2863K, the reaction zone duration measures 45 ms, representing a 15 ms reduction compared to the LTB model, while at 3001K, the reaction zone duration dwindles to a mere 25 ms. Simultaneously, benefiting from the high temperature fireball, the particle size distribution dimension of the reaction zone is widened, allowing aluminum powders with a larger particle size to undergo complete reaction within the secondary fireball. In accordance with the ascending order of fireball temperatures, the upper threshold for fully reacted aluminum powder particle size stands at 400 μm and 352 μm, respectively. The slightly higher upper limit in the size distribution dimension for TBX5% primarily results from the extended reaction zone. The degree of complete aluminum powder's reaction in both thermobaric explosives is 96.51% and 95.22%, respectively, demonstrating a fundamental similarity.

Another noteworthy observation is that, in the case of TBX5%, aluminum powders with a particle size below 16.4 μm exhibit a combustion duration of less than 1 ms (Eq. (2)). This particle size corresponds to the initial point on the burn-off time curve, whereas for TBX30%, this threshold is at 40.1 μm. It can be observed that a higher oxidizer content has the capability to generate a more elevated-temperature explosion fireball, wherein a substantial quantity of oxidizer decomposition byproducts disperses. In an extremely brief time frame, prior to the renewed ingress of fresh air into the reaction zone, this phenomenon leads to the rapid consumption of a significant quantity of aluminum powders, liberating a considerable amount of thermal energy. This energy supplementation accounts for the heat required to elevate aluminum powders to the temperature of explosive products and to facilitate aluminum powders gasification, thus sustaining the presence of a high-temperature fireball for an extended duration. Consequently, this study refers to the resultant effect as a stronger thermobaric effect.

5. Conclusion

This study designed and prepared distinct CL-20 based thermobaric explosives (TBX30% and TBX5%) alongside volumetrically equivalent TNT charges (RefTNT30% and RefTNT5%). Through static detonation tests, the mechanism of the explosion afterburning effect was investigated from multiple perspectives, and technical approaches for its enhancement were elucidated.

(1) While both thermobaric and TNT charges generated similarly expanding and deforming fireballs, their combustion mechanisms differed. The thermobaric explosive fireball showed a concentrated hotspot, confirming its volumetric explosion behavior, in stark contrast to the fragmented combustion observed in the TNT fireball.

(2) The temporal evolution of the fireball surface peak temperature differs significantly among the samples. TNT exhibits a brief increase followed by a decay, TBX30% shows a steady decline, while TBX5% maintains a plateau (15-45 ms) before decaying. Thermobaric explosives initially surpass same-volume TNT by over 1500 °C and sustain higher temperatures thereafter. Furthermore, a thermal crossover occurs between TBX5% and TBX30% at 30 ms, with their curves converging after 60 ms.

(3) All explosives exhibited afterburning, with TBX5% showing the most intense reaction, localizing its fireball hotspot within this zone. In contrast, TBX30%'s afterburning was weaker than TNT's, potentially due to its high oxidizer content. Furthermore, the confined fireball hotspots of thermobaric charges confirmed a volumetric explosion mechanism. The greater hotspot migration in TBX5% versus TBX30% was attributed to its lower oxidizer content, which results in a weaker initial blast field that cannot sustain immediate metal-air combustion. This indirectly verifies that higher oxidizer content promotes second-stage oxygen-deficient combustion while suppressing secondary afterburning.

(4) The post-combustion effect of thermobaric explosives also influences their blast wave parameters. Owing to the supplementary effect of post-combustion, TBX30% exhibits advantages such as higher impulse and longer positive pressure duration in the far field, whereas TBX5% demonstrates clear advantages in the near field.

(5) The combustion of aluminum powders in thermobaric explosives was assessed using two models. The LTB model, based on the alumina melting point, proved inaccurate, severely underestimating combustion extent and overestimating burn time. The superior AFT model, which uses an average fireball temperature, revealed high combustion completeness (>95%) for both TBX5% and TBX30%. It showed that higher oxidizer content in TBX30% resulted in a hotter fireball (3001K), leading to a shorter but more intense reaction that consumed finer aluminum particles faster, thereby enhancing the thermobaric output compared to TBX5%.

CRediT authorship contribution statement

Wen-xiang Bian: Writing – review & editing, Writing – original draft, Validation, Project administration, Data curation. **Hai-jun Zhao:** Resources, Data curation. **Jian Li:** Writing – review & editing, Validation, Resources. **Bo-liang Wang:** Writing – review & editing, Supervision. **Ya-ning Li:** Writing – review & editing, Writing – original draft, Validation, Project administration, Data curation.

Data availability

Data will be made available on request.

Conflict of interest

The authors declare the following financial interests which may be considered as potential competing interests: the author Hai-junZhao is currently employed by Ansteel Industrial Micro Fine Aluminum Powder Co., Ltd. The other authors declare that they have no competing interests.

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